Discourse on the Method of Rightly Conducting the Reason and Seeking Truth in the Field of Science

René Descartes

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Introduction

The cultural turn that symbolically and philosophically separates Aristotle/Medieval views from the "modern" era can be found in Descartes' writings. Born just before 1600, he witnessed and participated in the beginnings of modern science, especially with its empirical [The method of philosophical inquiry that holds that the human mind can only understand the world by beginning with the experiences of the senses.], mechanistic [The description of the motion of the universe in terms of the scientific notion of cause and effect alone. The image of the universe is that of a machine or mechanism that runs according to its construction and cannot deviate from it.] and mathematical under-girding. This Discourse on Method is, to some extent, an autobiography of an individual's evolution from the old system to the new.

This text was formally the first of his publications, around mid-1637, however it shows indications of being written perhaps ten years earlier. He claims he is not intending to reveal the "method" but only to "reveal how I have tried to direct my own [mind]." Much of the method is applied in the Meditations. But, here it is more historically positioned. It is important to remember that this text contains elements that were highly controversial at the time-views which even contributed to Galileo's house arrest under the Inquisition in 1633.

Commentary

In Part One, Descartes humbly begins by asserting that reason is "naturally equal among all men." He follows Aristotle by asserting that the differences, which make some seem much brighter than others, is a matter of degree, which is not essential to being human. He recounts his formal education from one of the finer schools in Europe, and indicates how he finds each element wanting: Languages, Fables, Books, Oratory, Poetry, Math, Morals, Jurisprudence and science. His disillusionment with formal education led him

to speculate that the average man on the street may know more, since his life was tied to it.

Part Two takes you further into his intellectual development. He observes how specialization, wherein many people work separately, does not always yield a higher quality product. In contrast, when a single person does the work, the product is often better. But, one need not demolish a whole city to rebuild anew. Descartes' project is much smaller.

Perhaps as a bow to his fears about the Church and the Inquisition, Descartes claims he is only working on his own mind, and is not trying to start a reformation. To avoid error, and build an edifice of knowledge he lays down four laws of thought: 1. Do not accept anything as true unless it was clear and distinct and about which he had no reason to doubt. 2. Break down difficult problems into easier problems. 3. Build carefully and slowly on the simplest and most easily known truths [For Descartes, truth is a matter of clear and distinct perceptions. Those ideas that are the product of clear and distinct mental perceptions are true.]. 4. To explain sufficiently so as to leave nothing out. Like geometry, which Descartes admired greatly, he sought strong and certain foundations on which to build a stable system of knowledge. (See his Meditations for more on this.)

In Part Three, he lays out a set of morals that he can use to guide his life, during the time in which he is undergoing this intellectual rebuilding. 1. Obey the laws and customs of the country. 2. Be firm and decisive in his actions. 3. Be master of himself and his desires, and not try to change the world (a rather ironic aim, looking back over the past four hundred years). 4. Choose the best occupation, which is to be a philosopher and scientist and following these new rules of reason.

Part Four recounts his Meditations in brief: how he lays the foundation with "I think, therefore I am," then builds on "clear and distinct ideas," all of which is then further supported by the necessary existence of God.

Part Five is essentially an overview of another of his works The World and Treatise on Man. Essentially he provides a mechanistic account of the world. He argues how God could have created the world with simple matter and some laws of motion, and "that by this means alone all purely material things could in the course of time have come to just as we now see them." Similarly, he gives an account of human anatomy, and especially the heart, relying heavily on the recently published and groundbreaking text on circulation and the heart by William Harvey in 1628. The laws of mechanics and motion, Descartes claims, can account for all of this. He concludes this section by comparing man to a machine and to an animal. In each case he argues why our rational soul indicates how we are not merely machines or animals.

Finally, in Part Six, Descartes discusses his reasoning with regard to not publishing this text when it was first written. His chief concern was with the Church's condemnation of Galileo's Dialogue (1632), which argued that the Earth revolves around the sun. The problem is that Descartes also thinks that some of his mechanics can benefit his fellow man, once more work is done with "these principles." As such, he feels a certain obligation to release it (at some time, at least). In the mean time, however, his intention is to continue working with the new method or path. He knows, for example, that he is merely one of many who will need to make ever-continuing observations to make more and more useful knowledge. He concludes the Discourse with a sort of declaration to continue his work, especially in applying his method to medicine.

Reading

If this discourse seems too long to be read at one sitting, it may be divided into six parts. In the first will be found various thoughts on the sciences; in the second, the principal rules of the method the author has used; in the third, some moral rules derived from this method; in the fourth, his proofs of the existence of God and of the human soul which form the basis of his philosophy; in the fifth are treated some questions of physics, especially the explanation of the heartbeat and of some other difficulties in medicine, as well as the difference between the souls of men and animals; and in the last, some prerequisites for further advances in the study of nature, as well as the author's reasons for writing this work.

PART ONE

SOME THOUGHTS ON THE SCIENCES

Good sense is mankind's most equitably divided endowment, for everyone thinks that he is so abundantly provided with it that 2 even those most difficult to please in other ways do not usually want more than they have of this. As it is not likely that everyone is mistaken, this evidence shows that the ability to judge correctly, and to distinguish the true from the false-which is really what is meant by good sense or reason-is the same by nature in all men; and that differences of opinion are not due to differences in intelligence, but merely to the fact that we use different approaches and consider different things. For it is not enough to have a good mind: one must use it well. The greatest souls are capable of the greatest vices as well as of the greatest virtues; and those who walk slowly can, if they follow the right path, go much farther than those who run rapidly in the wrong direction.

As for myself, I have never supposed that my mind was above the ordinary. On the contrary, I have often wished to have as quick a wit or as clear and distinct an imagination, or as ready and retentive a memory, as another person. And I know of no other qualities which make for a good mind, because as far as reason is concerned, it is the only thing which makes us men and distinguishes us from the animals, and I am therefore satisfied that it is fully present in each one of us. In this I follow the general opinion of philosophers, who say that there are differences in degree only in the 3 accidental qualities, and not in the essential qualities or natures of individuals of the same species.

But I do not hesitate to claim the good fortune of having stumbled, in my youth, upon certain paths which led me to certain considerations and maxims from which I formed a method of gradually increasing my knowledge and of improving my abilities as much as the mediocrity of my talents and the shortness of my life will permit. For I have already had such results that although in self-judgment I try to lean rather toward undervaluation than to presumption, I cannot escape a feeling of extreme satisfaction with the progress I believe I have already made in the search for truth [For Descartes, truth is a matter of clear and distinct perceptions. Those ideas that are the product of clear and distinct mental perceptions are true.]. And although from the philosophers' viewpoint almost all the activities of men appear to me as vain and useless, yet I conceive such hopes for the future that if some single one of the occupations of men, as men, should be truly good and important, I dare to believe that it is the one I have chosen.

It is always possible that I am wrong, and that I am mistaking a bit of copper and glass for gold and diamonds. I know bow subject we are to making false judgments in things that concern ourselves, and how much we ought to mistrust the judgments of our friends when they are in our own favor. But I should be glad to show in this Discourse what are the paths I have taken, and to present a sketch of my life, so that each one can form his own judgment of it. In this way I may learn from the opinions of those who read it, and thus add another to the methods of progress which I am accustomed to use. So it is not my intention to present a method which everyone ought to follow in order to think well, but only to show how I have made the attempt myself. Those who counsel others must consider themselves superior to those whom they counsel, and if they fall short in the least detail they are to blame. I only propose this writing as an autobiography, or, if you prefer, as a story in which you may possibly find some examples of conduct which you might see fit to imitate, as well as several others which you would have no reason to follow. I hope that it will prove useful to some without being harmful to any, and that all will take my frankness kindly.

From my childhood I lived in a world of books, and since I was taught that by their help I could gain a clear and assured knowledge of everything useful

in life, I was eager to learn from them. But as soon as I had finished the course of studies which usually admits one to the ranks of the learned, I changed my opinion completely. For I found myself saddled with so many doubts and errors that I seemed to have gained nothing in trying to educate myself unless it was to discover more and more fully how ignorant I was.

Nevertheless I had been in one of the most celebrated schools in Europe, where I thought there should be wise men if wise men existed anywhere on earth. I had learned there everything that others learned, and, not satisfied with merely the knowledge that was taught, I had perused as many books as I could find which contained more unusual and recondite knowledge. I also knew the opinions of others about myself, and that I was in no way judged inferior to my fellow students, even though several of them were preparing to become professors. And finally, it did not seem to me that our own times were less flourishing and fertile than were any of the earlier periods. All this led me to conclude that I could judge others by myself, and to decide that there was no such wisdom in the world as I had previously hoped to find.

I did not, however, cease to value the disciplines of the schools. I knew that the languages which one learns there are necessary to understand the works of the ancients; and that the delicacy of fiction enlivens the mind; that famous deeds of history ennoble it and, if read with understanding, aid in maturing one's judgment; that the reading of all the great books is like conversing with the best people of earlier times: it is even a studied conversation in which the authors show us only the best of their thoughts; that eloquence has incomparable powers and beauties; that poetry has enchanting delicacy and sweetness; that mathematics has very subtle processes which can serve as much to satisfy the inquiring mind as to aid all the arts and to diminish man's labor; that treatises on morals contain very useful teachings and exhortations to virtue; that theology teaches us how to go to heaven; that philosophy teaches us to talk with an appearance of truth about all things, and to make ourselves admired by the less learned; that law, medicine, and the other sciences bring honors and wealth to those who pursue them; and finally, that it is desirable to have examined all of them, even to the most superstitious and false, in order to recognize their real worth and avoid being deceived thereby.

But I thought that I had already spent enough time on languages, and even on reading the works of the ancients, and their histories and fiction. For conversing with the ancients is much like traveling. It is good to know something of the customs of various peoples, in order to judge our own more objectively, and so that we do not make the mistake of the untraveled in supposing that everything contrary to our customs is ridiculous and irrational. But when one spends too much time traveling, one becomes at last a stranger at home; and those who are too interested in things which occurred in past centuries are often remarkably ignorant of what is going on

today. In addition, fiction makes us imagine a number of events as possible which are really impossible, and even the most faithful histories, if they do not alter or embroider episodes to make them more worth reading, almost always omit the meanest and least illustrious circumstances so that the remainder is distorted. Thus it happens that those who regulate their behavior by the examples they find in books are apt to fall into the extravagances of the knights of romances, and undertake projects which it is beyond their ability to complete.

I esteemed eloquence highly, and loved poetry, but I felt that both were gifts of nature rather than fruits of study. Those who reason most cogently, and work over their thoughts to make them clear and intelligible, are always the most persuasive, even if they speak only a provincial dialect and have never studied rhetoric. Those who have the most agreeable imaginations and can express their thoughts with the most grace and color cannot fail to be the best poets, even if the poetic art is unknown to them.

I was especially pleased with mathematics, because of the certainty and self-evidence of its proofs; but I did not yet see its true usefulness and, thinking that it was good only for the mechanical arts, I was astonished that nothing more noble had been built on so firm and solid a foundation. On the other band, I compared the ethical writings of the ancient pagans to very superb and magnificent palaces built only on mud and sand: they laud the virtues and make them appear more desirable than anything else in the world; but they give no adequate criterion of virtue, and often what they call by such a name is nothing but apathy, parricide, pride or despair.

I revered our theology, and hoped as much as anyone else to get to heaven, but having learned on great authority that the road was just as open to the most ignorant as to the most learned, and that the truths of revelation which lead thereto are beyond our understanding, I would not have dared to submit them to the weakness of my reasonings. I thought that to succeed in their examination it would be necessary to have some extraordinary assistance from heaven, and to be more than a man.

I will say nothing of philosophy except that it has been studied for many centuries by the most outstanding minds without having produced anything which is not in dispute and consequently doubtful. I did not have enough presumption to hope to succeed better than the others; and when I noticed how many different opinions learned men may hold on the same subject, despite the fact that no more than one of them can ever be right, I resolved to consider almost as false any opinion which was merely plausible.

Finally, when it came to the other branches of learning, since they took their cardinal principles from philosophy, I judged that nothing solid could have been built on so insecure a foundation. Neither the honor nor the profit

to be gained thereby sufficed to make me study them, for I was fortunately not in such a financial condition as to make it necessary to trade upon my learning; and though I was not enough of a cynic to despise fame, I was little concerned with that which I could only obtain on false pretenses. And finally, I thought I knew enough of the disreputable doctrines not to be taken in by the promises of an alchemist, the predictions of an astrologer, the impostures of a magician, or by the tricks and boasts of any of those who profess to know that which they do not know.

This is why I gave up my studies entirely as soon as I reached the age when I was no longer under the control of my teachers. I resolved to seek no other knowledge than that which I might find within myself, or perhaps in the great book of nature. I spent a few years of my adolescence traveling, seeing courts and armies, living with people of diverse types and stations of life, acquiring varied experience, testing myself in the episodes which fortune sent me, and, above all, thinking about the things around me so that 1 could derive some profit from them. For it seemed to me that I might find much more of the truth in the cogitations which each man made on things which were important to him, and where he would be the loser if he judged badly, than in the cogitations of a man of letters in his study, concerned with speculations which produce no effect, and which have no consequences to him except perhaps that the farther they are removed from common sense, the more they titillate his vanity, since then he needs so much more wit and skill to make them seem plausible. Besides, I was always eager to learn to distinguish truth from falsehood, so that I could make intelligent decisions about the affairs of this life.

It is true that while I did nothing but observe the customs of other men, I found nothing there to satisfy me, and I noted just about as much difference of opinion as I had previously remarked among philosophers. The greatest profit to me was, therefore, that I became acquainted with customs generally approved and accepted by other great peoples that would appear extravagant and ridiculous among ourselves, and so I learned not to believe too firmly what I learned only from example and custom. Also I gradually freed myself from many errors which could obscure the light of nature and make us less capable of correct reasoning. But after spending several years in thus studying the book of nature and acquiring experience, I eventually reached the decision to study my own self, and to employ all my abilities to try to choose the right path. This produced much better results in my case, I think, than would have been produced if I had never left my books and my country.

PART TWO

THE PRINCIPAL RULES OF THE METHOD

I was then in Germany, where I had gone because of the wars which are still not ended; and while I was returning to the army from the coronation of the Emperor, I was caught by the onset of winter. There was no conversation to occupy me, and being untroubled by any cares or passions, I remained all day alone in a warm room. There I had plenty of leisure to examine my ideas. One of the first that occurred to me was that frequently there is less perfection in a work produced by several persons than in one produced by a single hand. Thus we notice that buildings conceived and completed by a single architect are usually more beautiful and better planned than those remodeled by several persons using ancient walls that had originally been built for quite other purposes. Similarly, those ancient towns which were originally nothing but hamlets, and in the course of time have become great cities, are ordinarily very badly arranged compared to one of the symmetrical metropolitan districts which a city planner has laid out on an open plain according to his own designs.

It is true that when we consider their buildings one by one, there is often as much beauty in the first city as in the second, or even more; nevertheless, when we observe how they are arranged, here a large unit, there a small; and how the streets are crooked and uneven, one would rather suppose that chance and not the decisions of rational men had so arranged them. And when we consider that there were always some officials in charge of private building, whose duty it was to see that they were conducive to the general good appearance of the city, we recognize that it is not easy to do a good job when using only the works of others. Similarly I supposed that peoples who were once half savage, and who became civilized by a gradual process and invented their laws one by one as the harmfulness of crimes and quarrels forced them to outlaw them, would be less well governed than those who have followed the constitutions of some prudent legislator from the time that their communities were founded. Thus it is quite certain that the condition of the true religion, whose rules were laid down by God alone, must be incomparably superior to all others. And, to speak of human affairs, I believe that Sparta was such a flourishing community, not because of the goodness of each of its laws in particular, seeing that many of them were very strange and even contrary to good morals, but because they were produced by a single legislator, and so all tended to the same end. And similarly I thought that the sciences found in books, at least those whose reasons were only probable and which had no proofs, have grown up little by little by the accumulation of the opinions of many different persons, and are therefore by no means as near to the truth as the simple and natural reasonings of a man of good sense concerning the things which he experiences. Likewise I thought that we were all children before being men, at which time we were necessarily under the control of our appetites and our teachers, and that neither of these influences is wholly consistent, and neither of them, perhaps, always tends toward the better. It is therefore impossible that our judgments should be as pure and

firm as they would have been had we the whole use of our reason from the time of our birth and if we had never been under any other control.

It is true that we never tear down all the houses in a city just to rebuild them in a different way and to make the streets more beautiful; but we do see that individual owners often have theirs torn down and rebuilt, and even that they may be forced to do so when the foundation is not firm and it is in danger of collapsing. By this example I was convinced that a private individual should not seek to reform a nation by changing all its customs and destroying it to construct it anew, nor to reform the body of knowledge or the system of education. Nevertheless, as far as the opinions which I had been receiving since my birth were concerned, I could not do better than to reject them completely for once in my lifetime, and to resume them afterwards, or perhaps accept better ones in their place, when I had determined how they fitted into a rational scheme. And I firmly believed that by this means I would succeed in conducting my life much better than if I built only upon the old foundations and gave credence to the principles which I had acquired in my childhood without ever having examined them to see whether they were true or not. For though I noticed several difficulties in the way, they were neither insurmountable nor comparable to those involved in the slightest reform of public affairs. For public affairs are on a large scale, and large edifices are too difficult to set up again once they have been thrown down, too difficult even to preserve once they have been shaken, and their fall is necessarily catastrophic. It is certain that many institutions have defects, since their differences alone guarantee that much, but custom has no doubt inured us to many of them. Custom has perhaps even found ways to avoid or correct more defects than prudence could have done. Finally, present institutions are practically always more tolerable than would be a change in them; just as highways which twist and turn among the mountains become gradually so easy to travel, as a result of much use, that it is much better to follow them than to attempt to go more directly by climbing cliffs and descending to the bottom of precipices.

That is why I cannot at all approve those mischievous spirits who, not being called either by birth or by attainments to a position of political power, are nevertheless constantly proposing some new reform. If I thought the slightest basis could be found in this Discourse for a suspicion that I was guilty of this folly, I would be loath to permit it to be published. Never has my intention been more than to try to reform my own ideas, and rebuild them on foundations that would be wholly mine. If my building has pleased me sufficiently to display a model of it to the public, it is not because I advise anyone to copy it. Those whom God has more bountifully endowed will no doubt have higher aims; there are others, I fear, for whom my own are too adventurous. Even the decision to abandon all one's preconceived notions is not an example for all to follow, and the world is largely composed of two sorts of individuals who should not try to follow it. First, there are those

who think themselves more able than they really are, and so make precipitate judgments and do not have enough patience to think matters through thoroughly. From this it follows that once they have taken the liberty of doubting their established principles, thus leaving the highway, they will never be able to keep to the narrow path which must be followed to go more directly, and will remain lost all their lives. Secondly, there are those who have enough sense or modesty to realize that they are less able to distinguish the true from the false than are others, and so should rather be satisfied to follow the opinions of these others than to search for better ones themselves.

As for myself, I should no doubt have belonged in the last class if I had had but a single teacher or if I had not known the differences which have always existed among the most learned. I had discovered in college that one cannot imagine anything so strange and unbelievable but that it has been upheld by some philosopher; and in my travels I had found that those who held opinions contrary to ours were neither barbarians nor savages, but that many of them were at least as reasonable as ourselves. I had considered how the same man, with the same capacity for reason, becomes different as a result of being brought up among Frenchmen or Germans than he would be if he had been brought up among Chinese or cannibals; and how, in our fashions, the thing which pleased us ten years ago and perhaps will please us again ten years in the future, now seems extravagant and ridiculous; and I felt that in all these ways we are much more greatly influenced by custom and example than by any certain knowledge. Faced with this divergence of opinion, I could not accept the testimony of the majority, for I thought it worthless as a proof of anything somewhat difficult to discover, since it is much more likely that a single man will have discovered it than a whole people. Nor, on the other hand, could I select anyone whose opinions seemed to me to be preferable to those of others, and I was thus constrained to embark on the investigation for myself.

Nevertheless, like a man who walks alone in the darkness, I resolved to go so slowly and circumspectly that if I did not get ahead very rapidly I was at least safe from falling. Also, I did not want to reject all the opinions which had slipped irrationally into my consciousness since birth, until I had first spent enough time planning how to accomplish the task which I was then undertaking, and seeking the true method of obtaining knowledge of everything which my mind was capable of understanding.

Among the branches of philosophy, I had, when younger, studied logic, and among those of mathematics, geometrical analysis and algebra; three arts or sciences which should be able to contribute something to my design. But in examining them I noticed that as far as logic was concerned its syllogisms and most of its other methods serve rather to explain to another what one already knows, or even, as the art of Lully, to speak without

judgment of what one does not know, than to learn new things. Although it does contain many true and good precepts, they are interspersed among so many others that are harmful or superfluous that it is almost as difficult to separate them as to bring forth a Diana or a Minerva from a block of virgin marble. Then, as far as the analysis of the Greeks and the algebra of the moderns is concerned, besides the fact that they deal with abstractions and appear to have no utility, the first is always so limited to the consideration of figures that it cannot exercise the understanding without greatly fatiguing the imagination, and the last is so limited to certain rules and certain numbers that it has become a confused and obscure art which perplexes the mind instead of a science which educates it. In consequence I thought that some other method must be found to combine the advantages of these three and to escape their faults. Finally, just as the multitude of laws frequently furnishes an excuse for vice, and a state is much better governed with a few laws which are strictly adhered to, so I thought that instead of the great number of precepts of which logic is composed, I would have enough with the four following ones, provided that 1 made a firm and unalterable resolution not to violate them even in a single instance.

The first rule was never to accept anything as true unless I recognized it to be evidently such: that is, carefully to avoid precipitation and prejudgment, and to include nothing in my conclusions unless it presented itself so clearly and distinctly to my mind that there was no occasion to doubt it.

The second was to divide each of the difficulties which .1 encountered into as many parts as possible, and as might be required for an easier solution.

The third was to think in an orderly fashion, beginning with the things which were simplest and easiest to understand, and gradually and by degrees reaching toward more complex knowledge, even treating as though ordered materials which were not necessarily so.

The last was always to make enumerations so complete, and reviews so general, that I would be certain that nothing was omitted.

Those long chains of reasoning, so simple and easy, which enabled the geometricians to reach the most difficult demonstrations, had made me wonder whether all things knowable to men might not fall into a similar logical sequence. If so, we need only refrain from accepting as true that which is not true, and carefully follow the order necessary to deduce each one from the others, and there cannot be any propositions so abstruse that we cannot prove them, or so recondite that we cannot discover them. It was not very difficult, either, to decide where we should look for a beginning, for I knew already that one begins with the simplest and easiest to know.

Considering that among all those who have previously sought truth in the sciences, mathematicians alone have been able to find some demonstrations, some certain and evident reasons, I had no doubt that I should begin where they did, although I expected no advantage except to accustom my mind to work with truths and not to be satisfied with bad reasoning. I do not mean that I intended to learn all the particular branches of mathematics; for I saw that although the objects they discuss are different, all these branches are in agreement in limiting their consideration to the relationships or proportions between their various objects. I judged therefore that it would be better to examine these proportions in general, and use particular objects as illustrations only in order to make their principles easier to comprehend, and to be able the more easily to apply them afterwards, without any forcing, to anything for which they would be suitable. I realized that in order to understand the principles of relationships I would sometimes have to consider them singly, and sometimes in groups. I thought I could consider them better singly as relationships between lines, because I could find nothing more simple or more easily pictured to my imagination and my senses. But in order to remember and understand them better when taken in groups, I had to express them in numbers, and in the smallest numbers possible. Thus I took the best traits of geometrical analysis and algebra, and corrected the faults of one by the other.

The exact observation of the few precepts which I had chosen gave me such facility in clarifying all the issues in these two sciences that it took only two or three months to examine them. I began with the most simple and general, and each truth that I found was a rule which helped me to find others, so that I not only solved many problems which I had previously judged very difficult, but also it seemed to me that toward the end I could determine to what extent a stillunsolved problem could be solved, and what procedures should be used in solving it. In this I trust that I shall not appear too vain, considering that there is only one true solution to a given problem, and whoever finds it knows all that anyone can know about it. Thus, for example, a child who has learned arithmetic and had performed an addition according to the rules may feel certain that as far as that particular sum is concerned, he has found everything that a human mind can discover. For, after all, the method of following the correct order and stating precisely all the circumstances of what we are investigating is the whole of what gives certainty to the rules of arithmetic.

What pleased me most about this method was that it enabled me to reason in all things, if not perfectly, at least as well as was in my power. In addition, I felt that in practicing it my mind was gradually becoming accustomed to conceive its objects more clearly and distinctly, and since I had not directed this method to any particular subject matter, I was in hopes of applying it just as usefully to the difficulties of other sciences as I had already to those of algebra. Not that I would dare to undertake to

examine at once all the difficulties that presented themselves, for that would have been contrary to the principle of order. But I had observed that all the basic principles of the sciences were taken from philosophy, which itself had no certain ones. It therefore seemed that I should first attempt to establish philosophic principles, and that since this was the most important thing in the world and the place where precipitation and prejudgment were most to be feared, I should not attempt to reach conclusions until I had attained a much more mature age than my then twentythree years, and had spent much time in preparing for it. This preparation would consist partly in freeing my mind from the false opinions which I had previously acquired, partly in building up a fund of experiences which should serve afterwards as the raw material of my reasoning, and partly in training myself in the method which I had determined upon, so that I should become more and more adept in its use.

PART THREE

SOME MORAL RULES DERIVED FROM THE METHOD

In planning to rebuild one's house it is not enough to draw up the plans for the new dwelling, tear down the old one, and provide materials and obtain workmen for the task. We must see that we are provided with a comfortable place to stay while the work of rebuilding is going on. Similarly in my own case; while reason obliged me to be irresolute in my beliefs, there was no reason why I should be so in my actions. In order to live as happily as possible during the interval I prepared a provisional code of morality for myself, consisting of three or four maxims which I here set forth.

The first was to obey the laws and customs of my country, constantly retaining the religion in which, by God's grace, I had been brought up since childhood, and in all other matters to follow the most moderate and least excessive opinions to be found in the practices of the more judicious part of the community in which I would live. For I was then about to discard my own opinions in order to reexamine them, and meanwhile could do no better than to follow those of the most reliable judges. While there may be, no doubt, just as reliable persons among the Persians or the Chinese as among ourselves, it seemed more practical to pattern my conduct on that of the society in which I would have to live. Furthermore, it seemed to me that to learn people's true opinions, I should pay attention to their conduct rather than to their words, not only because in our corrupt times there are few who are ready to say all that they believe, but also because many are not aware of their own beliefs, since the mental process of knowing a thing is distinct from, and can occur without, the mental process of knowing that we know it. Among a number of opinions equally widely accepted, I chose only the most moderate, partly because these are always the most convenient in

practice and, since excess is usually bad, presumably the best; but also so that I should stray a shorter distance from the true road in case I should make a mistake, than I would in choosing one extreme when it was the other that should have been followed. In particular, I considered as excessive all the promises by which we abandon some of our freedom. Not that I disapproved of the laws which, to remedy the inconstancy of vacillating spirits, permit them to make bonds or contracts which oblige them to persevere with their intentions, provided the intentions are good, or at least not bad, but because I recognized that nothing is unchanging, and that in my own case I was proposing to improve my judgment more and more, not to make it worse. It would therefore have been a major violation of common sense if I obliged my. self to continue to accept a thing I formerly approved after it ceased to merit approval, or after I altered my opinion of it.

My second maxim was to be as firm and determined in my actions as I could be, and not to act on the most doubtful decisions, once I had made them, any less resolutely than on the most certain. In this matter I patterned my behavior on that of travelers, who, finding themselves lost in a forest, must not wander about, now turning this way, now that, and still less should remain in one place, but should go as straight as they can in the direction they first select and not change the direction except for the strongest reasons. By this method, even if the direction was chosen at random, they will presumably arrive at some destination, not perhaps where they would like to be, but at least where they will be better off than in the middle of the forest. Similarly, situations in life often permit no delay; and when we cannot determine the course which is certainly best, we must follow the one which is probably the best; and when we cannot determine even that, we must nevertheless select one and follow it thereafter as though it were certainly best. If the course selected is not indeed a good one, at least the reasons for selecting it are excellent. This frame of mind freed me also from the repentance and remorse commonly felt by those vacillating individuals who are always seeking as worth while things which they later judge to be bad.

My third maxim was always to seek to conquer myself rather than fortune, to change my desires rather than the established order, and generally to believe that nothing except our thoughts is wholly under our control, so that after we have done our best in external matters, what remains to be done is absolutely impossible, at least as far as we are concerned. This maxim in itself should suffice to prevent me from desiring in the future anything which I could not acquire, and thus to make me happy. For it is our nature to desire only that which we imagine to be somehow attainable, and if we consider all external benefits equally beyond our reach we will no more regret being unjustly deprived of our birthright than we regret not possessing the kingdoms of China or Mexico. Thus, making a virtue of necessity, we no more desire to be well when we are sick, or to be free when we are in prison, than we now desire bodies as incorruptible as diamonds, or

wings to fly like the birds. But I must admit that it takes much practice and frequently repeated meditations to become accustomed to view things in this manner, and I think that this must have been the principal secret of those philosophers of ancient times who were able to rise above fortune, and, despite pains and poverty, to vie with the gods in happiness. Being constantly occupied in considering the limits imposed upon them by nature, they were so perfectly convinced that nothing was really theirs but their thoughts that that alone was sufficient to keep them from any concern in other things. Their control of their thoughts, on the other hand, was so absolute that they had some justification for considering themselves richer and more powerful, more free and happier, than any other man who did not have this philosophy, and who, however much he might be favored by nature and fortune, had no such control over his desires.

Finally, I planned to make a review of the various occupations possible in this life, in order to choose the best. Without intending to disparage other occupations, I thought I could do no better than to continue in the one I was engaged in, employing my life in improving my mind and increasing as far as I could my knowledge of the truth by following the method that I had outlined for myself. I had experienced such periods of great happiness after I had begun to use this method, that I could hope for no greater or more innocent joys in this life. In discovering day after day truths which seemed fairly important and generally unknown to other men, I was filled with such satisfaction that other considerations did not affect me. Another reason for my decision was that the three maxims previously considered were based on my plan to continue the search for truth. For as God has given each one of us some ability to distinguish the true from the false, I should not have been content for one instant to rely on the opinions of others if I had not planned to use my own judgment at the proper time; nor could I have followed those opinions with a clear conscience if I had not hoped to take advantage of every opportunity to find better ones, if better ones there were. And finally, I could not have limited my desires, nor been happy, if I were not following a path by which I expected to obtain all the knowledge of which I was capable and, by the same token, all the real values to which I might aspire. Besides, since our will neither seeks nor avoids anything except as it is judged good or bad by our reason, good judgment is sufficient to guarantee good behavior. Judging as best one can therefore implies that one acts as well as one can, or in other words, that one will acquire all the virtues and with them all other possible goods. Once we are sure of this, we cannot well fail to be happy.

After thus assuring myself of these maxims, and having put them aside with the truths of the Faith, which have always been most certain to me, I judged that I could proceed freely to reject all my other beliefs. And inasmuch as I hoped to obtain my end more readily by conversing with men than by remaining any longer in my warm retreat, where I had had all these

thoughts, I proceeded on my way before winter was wholly passed. In the nine years that followed I wandered here and there throughout the world, trying everywhere to be spectator rather than actor in all the comedies that go on. I took particular pains in judging each thing to seek out whatever elements of uncertainty it contained, which might cause us to conceive false opinions about it. Meanwhile I tried to clear my mind of all the errors that had previously accumulated. In this I did not wish to imitate the sceptics, who doubted only for the sake of doubting and intended to remain always irresolute; on the contrary, my whole purpose was to achieve greater certainty and to reject the loose earth and sand in favor of rock and clay.

In all these things I seemed to succeed well enough, for, as I was trying to discover the falsity or uncertainty of the propositions I was examining, not by feeble conjectures but by clear and assured reasonings, I encountered nothing that did not lead me to some certain conclusions, even if it were only that the matter was wholly uncertain. And just as in tearing down a building we usually retain the debris to help build a new one, so in destroying all of my opinions which seemed to me ill-founded, I made many observations and acquired much experience which has since aided me in establishing more certain knowledge. In addition, I continued to practice the method which I had decided upon; and besides conducting all my thoughts according to its rules, I set aside a few hours now and then for practice upon mathematical difficulties. In some cases I even practiced upon some other difficulties which could be made to parallel mathematical ones by rejecting those principles of the sciences in question which I did not find sufficiently well established, as I have explained in some of my other writings. Thus I lived, in appearance, just like those who have nothing to do but to live a pleasant and innocent life and attempt to obtain the pleasures without the vices, to enjoy their leisure without ennui, and to occupy their time with all the respectable amusements available. But in reality I never desisted from my design and continued to achieve greater acquaintance with truth, perhaps more than I would have if I bad only read books or sought the society of men of letters.

In any case, nine years passed before I reached my decision about the difficulties ordinarily in dispute among the learned, and before I sought to lay the groundwork of a philosophy more certain than popular belief. The example of several men of excellent abilities who had previously attempted my task and who, in my opinion, had failed, made me fear so many difficulties that I should perhaps not have dared to start so soon if I had not learned of a rumor that I had already completed my philosophy. I did not know on what such an opinion was based; if I contributed somewhat to it by my conversation, it must have been by confessing my ignorance more freely than is usually the case among those who have studied a little, and possibly also by presenting my reasons for doubting many things that others deemed certain. I am sure that I did not boast of any doctrines. But I did not want to

be taken for more than I was, and so I thought that I should try by all means to make myself worthy of my reputation. Just eight years ago, therefore, I. decided to abandon those places where I would be among acquaintances, and retired to Holland, where the long duration of the war produced such conditions that the armies billeted there seemed but to guarantee the fruits of peace. There, in the midst of a great and busy people, more interested in their own affairs than curious about those of others, I was able to enjoy all the comforts of life to be found in the most populous cities while living in as solitary and retired a fashion as though in the most remote of deserts.

PART FOUR

PROOFS OF THE EXISTENCE OF GOD AND OF THE HUMAN SOUL

I do not know whether I ought to touch upon my first meditations here, for they are so metaphysical and out of the ordinary that they might not be interesting to most people. Nevertheless, in order to show whether my fundamental notions are sufficiently sound, I find myself more or less constrained to speak of them. I had noticed for a long time that in practice it is sometimes necessary to follow opinions which we know to be very uncertain, just as though they were indubitable, as I stated before; but inasmuch as I desired to devote myself wholly to the search for truth, I thought that I should take a course precisely contrary, and reject as absolutely false anything of which I could have the least doubt, in order to see whether anything would be left after this procedure which could be called wholly certain. Thus, as our senses deceive us at times, I was ready to suppose that nothing was at all the way our senses represented them to be. As there are men who make mistakes in reasoning even on the simplest topics in geometry, I judged that I was as liable to error as any other, and rejected as false all the reasoning which I had previously accepted as valid demonstration. Finally, as the same precepts which we have when awake may come to us when asleep without their being true, I decided to suppose that nothing that had ever entered my mind was more real than the illusions of my dreams. But I soon noticed that while I thus wished to think everything false, it was necessarily true that I who thought so was something. Since this truth, I think, therefore I am,, was so firm and assured that all the most extravagant suppositions of the sceptics were unable to shake it, I judged that I could safely accept it as the first principle of the philosophy I was seeking.

I then examined closely what I was, and saw that I could imagine that I had no body, and that there was no world nor any place that I occupied, but that I could not imagine for a moment that I did not exist. On the contrary, from the very fact that I doubted the truth of other things, it followed very evidently and very certainly that I existed. On the other hand, if I had ceased

to think while all the rest of what I had ever imagined remained true, I would have had no reason to believe that I existed; therefore I concluded that I was a substance whose whole essence or nature was only to think, and which, to exist, has no need of space nor of any material thing. Thus it follows that this ego, this soul, by which I am what I am, is entirely distinct from the body and is easier to know than the latter, and that even if the body were not, the soul would not cease to be all that it now is.

Next I considered in general what is required of a proposition for it to be true and certain, for since I had just discovered one to be such, I thought I ought also to know of what that certitude consisted. I saw that there was nothing at all in this statement, "I think, therefore I am," to assure me that I was saying the truth, unless it was that I saw very clearly that to think one must exist. So I judged that I could accept as a general rule that the things which we conceive very clearly and very distinctly are always true, but that there may well be some difficulty in deciding which are those which we conceive distinctly.

After that I reflected upon the fact that I doubted, and that, in consequence, my spirit was not wholly perfect, for I saw clearly that it was a greater perfection to know than to doubt. I decided to ascertain from what source I had learned to think of something more perfect than myself, and it appeared evident that it must have been from some nature which was in fact more perfect. As for my ideas about many other things outside of me, as the sky, earth, light, heat, and thousands of other things, I was not so much troubled to discover where they came from, because I found nothing in them superior to my own nature. If they really existed, I could believe that whatever perfection they possessed might be derived from my own nature; if they did not exist, I could believe that they were derived from nothingness, that is, that they were derived from my own defects. But this could not be the explanation of my idea of a being more perfect than my own. To derive it from nothingness was manifestly impossible, and it is no less repugnant to good sense to assume what is more perfect comes from and depends on the less perfect than it is to assume that something comes from nothing, so that I could not assume that it came from my. self. Thus the only hypothesis left was that this idea was put in my mind by a nature that was really more perfect than I was, which had all the perfections that I could imagine, and which was, in a word, God. To this I added that since I knew some perfections which I did not possess, I was not the only being in existence (I will here use freely, if you will pardon me, the terms of the schools), and that it followed of necessity that there was someone else more perfect upon whom I depended and from whom I had acquired all that I possessed. For if I had been alone and independent of anything else, so that I had bestowed upon myself all that limited quantity of value which I shared with the perfect Being, I would have been able to get from myself, in the same way, all the surplus which I recognize as lacking in me, and so would

have been myself infinite, eternal, immutable, omniscient, omnipotent, and, in sum, I would possess all the perfections that I could discover in God.

For, following the reasoning which I have just explained, to know the nature of God as far as I was capable of such knowledge, I had only to consider each quality of which I had an idea, and decide whether it was or was not a perfection to possess it. I would then be certain that none of those which had some imperfection was in him, but that all the others were. I saw that doubt, inconstancy, sorrow and similar things could not be part of God's nature, since I would be happy to be without them myself. In addition, I had ideas of many sensible and corporeal entities, for although I might suppose that I was dreaming and that all that I saw or imagined was false, I could not at any rate deny that the ideas were truly in my consciousness. Since I had already recognized very clearly that intelligent nature is distinct from corporeal nature, I considered that composition is an evidence of dependency and that dependency is manifestly a defect. From this I judged that it could not be a perfection in God to be composed of these two natures, and that consequently he was not so composed. But if there were in the world bodies, or even intelligences or other natures that were not wholly perfect, their being must depend on God's power in such a way that they could not subsist without him for a single moment.

At this point I wished to seek for other truths, and proposed for consideration the object of the geometricians. This I conceived as a continuous body, or a space infinitely extended in length, breadth, and height or depth; divisible into various parts which can have different shapes and sizes and can be moved or transposed in any way: all of which is presumed by geometricians to be true of their object. I went through some of their simplest demonstrations and noticed that the great certainty which everyone attributes to them is only based on the fact that they are evidently conceived, following the rule previously established. I noticed also that there was nothing at all in them to assure me of the existence of their object; it was clear, for example, that if we posit a triangle, its three angles must be equal to two right angles, but there was nothing in that to assure me that there was a single triangle in the world. When I turned back to my idea of a perfect Being, on the other hand, I discovered that existence was included in that idea in the same way that the idea of a triangle contains the equality of its angles to two right angles, or that the idea of a sphere includes the equidistance of all its parts from its center. Perhaps, in fact, the existence of the perfect Being is even more evident. Consequently, it is at least as certain that God, who is this perfect Being, exists, as any theorem of geometry could possibly be. What makes many people feel that it is difficult to know of the existence of God, or even of the nature of their own souls, is that they never consider things higher than corporeal objects. They are so accustomed never to think of anything without picturing it-a method of thinking suitable only for material objects-that everything which

is not picturable seems to them unintelligible. This is also manifest in the fact that even philosophers hold it as a maxim in the schools that there is nothing in the understanding which was not first in the senses, a location where it is clearly evident that the ideas of God and of the soul have never been. It seems to me that those who wish to use imagery to understand these matters are doing precisely the same thing that they would be doing if they tried to use their eyes to hear sounds or smell odors. There is even this difference: that the sense of sight gives us no less certainty of the truth of objects than do those of smell and hearing, while neither our imagery nor our senses could assure us of anything without the cooperation of our understanding.

Finally, if there are still some men who are not sufficiently persuaded of the existence of God and of their souls by the reasons which I have given, I want them to understand that all the other things of which they might think themselves more certain, such as their having a body, or the existence of stars and of an earth, and other such things, are less certain. For even though we have a moral assurance of these things, such that it seems we cannot doubt them without extravagance, yet without being unreasonable we cannot deny that, as far as metaphysical certainty goes, there is sufficient room for doubt. For we can imagine, when asleep, that we have another body and see other stars and another earth without there being any such. How could one know that the thoughts which come to us in dreams are false rather than the others, since they are often no less vivid and detailed? Let the best minds study this question as long as they wish, I do not believe they can find any reason good enough to remove this doubt unless they presuppose the existence of God. The very principle which I took as a rule to start with, namely, that all those things which we conceived very clearly and very distinctly are true, is known to be true only because God exists, and because he is a perfect Being, and because everything in us comes from him. From this it follows that our ideas or notions, being real things which come from God insofar as they are clear and distinct, cannot to that extent fail to be true. Consequently, though we often have ideas which contain falsity, they can only be those ideas which contain some confusion and obscurity, in which respect they participate in nothingness. That is to say, they are confused in us only because we are not wholly perfect. It is evident that it is no less repugnant to good sense to assume that falsity or imperfection as such is derived from God, as that truth or perfection is derived from nothingness. But if we did not know that all reality and truth within us came from a perfect and infinite Being, however clear and distinct our ideas might be, we would have no reason to be certain that they were endowed with the perfection of being true.

After the knowledge of God and the soul has thus made us certain of our rule, it is easy to see that the dreams which we have when asleep do not in any way cast doubt upon the truth of our waking thoughts. For if it

happened that we had some very distinct idea, even while sleeping, as for example when a geometrician dreams of some new proof, his sleep does not keep the proof from being good. As for the most common error of dreams, which is to picture various objects in the same way as our external senses represent them to us, it does not matter if this gives us a reason to distrust the truth of the impressions we receive from the senses, because we can also be mistaken in them frequently without being asleep, as when jaundiced persons see everything yellow, or as the stars and other distant objects appear much smaller than they really are. For in truth, whether we are asleep or awake, we should never allow ourselves to be convinced except on the evidence of our reason. Note that I say of our reason,, and not of our imagination or of our senses; for even though we see the sun very clearly, we must not judge thereby that its size is such as we see it, and we can well imagine distinctly the head of a lion mounted on the body of a goat, without concluding that a chimera exists in this world. For reason does not insist that all we see or visualize in this way is true, but it does insist that all our ideas or notions must have some foundation in truth, for it would not be possible that God, who is allperfect and wholly truthful, would otherwise have given them to us. Since our reasonings are never as evident or as complete in sleep as in waking life, although sometimes our imaginations are then as lively and detailed as when awake, or even more so, and since reason tells us also that all our thoughts cannot be true, as we are not wholly perfect; whatever of truth is to be found in our ideas will inevitably occur in those which we have when awake rather than in our dreams.

PART FIVE

SOME QUESTIONS OF PHYSICS

I would have been glad to continue my exposition and exhibit here the whole chain of other truths which I deduced from these basic ones, but for the fact that to do so I should have to speak of many questions which are in dispute among the learned. As I do not wish to embroil myself with them, I think it would be better to abstain, so I shall give only an outline of these views, and let wiser people judge whether the public should be informed in greater detail.

I have always remained true to the resolution I made, never to suppose any other principle than that which I have just used to demonstrate the existence of God and the soul, and not to admit anything as true which did not seem to me clearer and more certain than the demonstrations of the geometricians previously seemed Nevertheless, I have not only succeeded in satisfying myself in this short time on all the principal difficulties usually treated in philosophy, but have also discovered certain laws which God has so established in nature, and the notion of which he has so fixed in our

minds, that after sufficient reflection we cannot doubt that they are exactly observed in all which exists or which happens in the world. Finally, in considering the implications of these laws I seem to have discovered several truths more useful and more important than anything I had previously learned or even hoped to learn.

Since I have tried to explain the most important of these laws in a work which certain considerations prevent my publishing, I see no better way to proceed than by summarizing its contents. I intended to include in it all that I thought I knew, before writing it, concerning the nature of material things. But I found myself in the same state as painters, who cannot equally well represent in a twodimensional painting all the various faces of a solid body, and so choose one to bring to the light, and leave the others in shadow, so that they can be seen only while viewing the selected side. Therefore, fearing that I would not be able to put into any discourse all that I intended, I undertook solely to describe at length what I thought on the subject of light, and took that occasion to add something concerning the sun and the fixed stars, since they are almost the only sources of light; of the sky, since it transmits it; of the planets, the comets, and the earth, since they reflect it; and in particular of all the objects on earth, since they are either colored or transparent or luminous; and finally of man, since he is the observer of it. I even elected, as a painter might do, to place my object somewhat in the shadow, so that I could express my opinions more freely without being obliged to accept or to refute the opinions commonly held by the learned. therefore resolved to leave this world for them to dispute about, and to speak only of what would happen in a new one, if God should now create, somewhere in imaginary space, enough matter to make one; and if he agitated the various parts of this matter without order, making a chaos as confused as the poets could imagine, but that afterward he did nothing but lend his usual support to nature, allowing it to behave according to the laws he had established.

So I first described this matter and tried to picture it in such a way that nothing in the world could be clearer or more intelligible except what has just been said about God and the soul. I even expressly supposed that this matter had none of the forms or qualities concerning which one disputes in the schools, nor in general anything that we do not know so naturally that we cannot even pretend to ignore it. Furthermore, I showed what were the laws of nature, and without basing my reasons on anything more specific than the infinite perfections of God, I tried to demonstrate everything which might be doubtful, and to show that nature is such that even if God had created several worlds, there would have been none where these laws were not observed. After that, I showed how the greater part of the matter in this chaos would, in consequence of these laws, become arranged in a manner which would make it similar to our skies; and how nevertheless some of the parts must compose an earth, and some planets and comets, and

others a sun and fixed stars. And here, enlarging upon the topic of light, I explained at considerable length its nature when contained in the sun and the stars, how from there it traverses in an instant the immense reaches of the heavens, and how it is reflected from the planets and comets toward the earth. I also added several things concerning the substance, situation, movements, and all the diverse qualities of these celestial objects and stars; until I thought I had said enough to show that there were no phenomena of this world which would not or at least could not occur similarly in the world I was describing.

Thence I went on to speak particularly of the earth: how, even though I had expressly supposed that God had given no weight to the matter of which it was composed, all its parts would tend exactly toward its center; how the disposition of the celestial bodies and stars, principally the moon, would cause an ebb and flow in the water and air on its surface, similar in all respects to the tides of our seas, and in addition a certain current, as much of water as of air, from east to west, such as we find in the tropics; how mountains, seas, springs and rivers could naturally occur, metals come to be in mines, plants grow in the fields, and, in general, how the whole genus of mixed or composite objects would be formed.

Among other things, since outside of the stars I knew nothing but fire which produced light, I strove to explain quite dearly the whole of the nature of fire: how it is produced and maintained; how sometimes it has heat without light and sometimes light without heat; how it can produce different colors in different objects, and many other qualities; bow it melts some objects and hardens others; how it can consume things entirely or convert them into ashes and smoke; and finally, how by the violence of its action it turns ashes into glass; for as this transmutation of ashes into glass seemed as admirable as any that occurs in nature, I found a particular pleasure in describing it.

I did not wish to infer from all this that the world had been created in the manner I proposed, for it is much more likely that God created it in the beginning in the form it was to assume. But it is certain, and this is an opinion commonly held by theologians, that the action by which the world is now conserved is precisely the same as that by which it was created. Even therefore, if God had given the world in the beginning no other form but chaos, and had only established the laws of nature and given his concurrence for the world to behave as it usually does, one can believe, without injustice to the miracle of creation, that all material objects could have become, in time, such as we see them at present. Their nature is much easier to conceive when one pictures their gradual growth in this manner rather than considering them as produced in their completed state.

From the description of inanimate objects and plants I passed to that of

animals, and particularly of man. But I did not as yet know enough to speak of these in the same style as of the rest, in showing the causes of their existence and showing from what origin and in what manner nature must have produced them. I was therefore satisfied to assume that God formed the body of a man just like our own, both in the external configuration of its members and in the internal configuration of its organs, without using in its composition any matter but that which I had described. I also assumed that God did not put into this body, to start with, any rational soul or any other entity to serve as a vegetable or sensitive soul, beyond kindling in the heart one of those fires without light which I had already described and which I considered to be entirely similar to that which heats grain when it is stored before it is dry, or which warms new wines when they are allowed to ferment before being separated from the grapes. Examining the functions which such a body would have, I discovered everything that can exist without thinking; everything except that which is contributed by the soul: that part of us distinct from the body whose essence, as we have previously said, is only to think. These functions are the same as those in which the unreasoning animals resemble us, and do not include any of those which are dependent on thinking and which belong to us as men. These human qualities I discovered somewhat later, when I supposed that God created a rational soul and joined it to the body in a certain fashion which I described.

In order to show how I treated this matter, I wish to insert here the explanation of the function of the heart and arteries. As the first and most general function found in animals, it will serve to indicate what the reader should think of all the rest. Those who are not well versed in anatomy will find less difficulty in understanding what I am going to say if they will take the trouble, before reading this, to have the heart of some large animal cut open before them, for the heart of an animal with lungs is quite similar to that of man. Let them then observe the two chambers, or ventricles, which it contains. First, the one on the right side connects with two very large tubes: the vena cava, which is the principal container of blood and resembles the trunk of a tree of which all the other veins are branches; and the vena arteriosa, misnamed since it is really an artery which starts in the heart, then divides into several branches, and spreads throughout the lungs. The left ventricle has two similar tubes at least as large as those just described: the arteria venosa, likewise misnamed since it is purely a vein, coming from the lungs, where it is divided into a number of branches interlaced with those of the vena arteriosa and those of the windpipe, through which enters the air we breathe; and the aorta, which, starting from the heart, sends its branches everywhere throughout the body. I should suggest also that the reader ob. serve the eleven little membranes which, like so many little valves, open and close the four openings in these two ventricles. Three are at the entrance to the vena cava, where they are so disposed that they cannot stop the blood that it contains from flowing into the right ventricle but prevent any of it from flowing back. Three, at the entrance of the vena

arteriosa, are disposed in precisely the opposite fashion, permitting the blood in the heart to pass to the lungs but not allowing the blood in the lungs to return. So also there are two at the entrance to the arteria venosa, which permit the blood to pass from the lungs to the left ventricle and prevent its return, and three at the entrance to the aorta, permitting the blood to leave the heart but not to return. There is no need to seek any other reason for the number of these membranes other than the fact that the opening of the arteria venosa, being oval because of its location, can be conveniently closed with two, while the others are round and can be more readily closed with three. I should like the reader to notice also that the aorta and the vena arteriosa are much harder and firmer than the arteria venosa and vena cava, and that these last two are enlarged near the heart, forming two sacs called the "ears" or auricles of the heart, composed of a flesh resembling that of the ears. Notice also that there is always more heat in the heart than in any other part of the body, and that this heat is capable of causing any drop of blood which enters the ventricles to expand immediately, just as any liquid does when it falls drop by drop into some very hot vessel.

After that, I need say nothing more to explain the functioning of the heart, except that when its ventricles are not full of blood, some necessarily flows into it. The right ventricle is filled from the vena cava, and the left ventricle from the arteria venosa, since these two vessels are always full and their entrances, opening toward the heart, cannot then be closed. The portions of blood in each ventride cannot fail to be very large since the openings are very large and the vessels from which they came full of blood; and as soon as these portions enter the heart, they become rarefied and expand because of the heat there. This dilates the whole heart and pushes upon and closes the five valules at the entrance of the two vessels from which the blood comes, preventing any more blood from entering the heart. As the blood continues to expand, it pushes upon and opens the six other valvules which are at the entrance to the other two vessels through which the blood leaves, and thus inflates all the branches of the vena arteriosa and the aorta at almost the same instant as the heart. A moment later the heart and these arteries are all deflated because the blood which has entered them has cooled. The six valvules close, and the five of the vena cava and <i.>arteria venosa reopen permitting two more portions of blood to enter and dilate the heart and the arteries the same as before. And since the blood which thus enters the heart comes through the two sacs called auricles, it follows that the condition of the latter is the opposite of that of the ventricles, and that the former are deflated when the latter are inflated.

For the rest, so that those who do not appreciate the force of mathematical demonstration and are not accustomed to distinguish between good and bad reasons should not make the mistake of denying this without

examining it, I must warn them that the motion which I have just explained follows necessarily from the mere disposition of the parts of the heart visible to the naked eye, from the heat which one can feel with the fingers, and from the nature of the blood, which one can learn by experiment: just as the motions of a clock follow from the weight, location, and configuration of its counterweights and wheels.

But if one asks why the blood in the veins does not become exhausted by thus flowing continually into the heart, and why the arteries do not become overfull, since all that passes through the heart goes there, I need only point out what has already been written by an English doctor who has the glory of having broken the ice in this matter. He 1 was the first to show that there are many small passages at the ends of the arteries, by which the blood received from the heart enters into the small branches of the veins, whence it returns again to the heart; so that its path is nothing but a perpetual circulation. This he proved very adequately by the common experience of surgeons, who, having applied a tourniquet to the arm, not too tightly, above the spot where they open a vein, make the blood flow more abundantly than it would without the tourniquet. On the other hand, quite the contrary occurs if they tie it below, between the hand and the opening, or even if they tie it very tightly above. For it is obvious that a tourniquet which is moderately tight can prevent the blood which is already in the arm from returning toward the heart through the veins, but cannot hinder that which is continually coming from the heart through the arteries, because the arteries are situated below the veins, and because their walls are stiffer and less easy to compress, and also because the blood comes from the heart with greater pressure than it has when returning through the veins. Since this blood leaves the arm through an opening in one of the veins, there must necessarily be some passages below the tourniquet, that is, toward the hand, by which it comes from the arteries. He also proves his contention about the circulation of the blood by certain small membranes so disposed in various places along the veins that they do not permit the blood to flow from the middle of the body toward the extremities, but only to return from the extremities toward the heart; and further by the fact that all the blood in the body can be lost in a very short time when a single artery is cut, even if it is tightly constricted close to the heart and cut between the constriction and the heart, so that there is no imaginable way that the blood which escapes comes from another source than the heart.

There are several other considerations which prove that the real cause of this movement of the blood is the one which I have given; such as, first, the difference between that which comes from the veins and that which comes from the arteries: a difference only to be explained by the fact that the blood is rarefied, as though it were distilled, in passing through the heart, and is therefore thinner, more active, and warmer when it has just come from there and is in the arteries, than just before it enters and is in

the veins. Careful observation shows that this difference is more apparent near the heart, and is not so noticeable at points far removed from it. Then the hardness of the membranes of which the v arteriosa and the aorta are composed shows well enough that the blood passes through them with greater pressure than through the veins. Furthermore, why should the left ventricle and the aorta be larger and broader than the right ventricle and the vena arteriosa , if it is not that the blood of the arteria venosa, not having been in the lungs since it passed through the heart, is thinner and becomes more rarefied more readily than that which comes directly from the vena cava? And what could doctors tell by feeling the pulse, if they did not know that as the nature of the blood changes, it can be rarefied by the heat of the heart more or less strongly and more or less rapidly than before? And if we examine how this heat is communicated to the other parts of the body, must we not admit that it is through the blood which is warmed in passing through the heart and spreads this heat through the whole body? From this it results that if the blood is withdrawn from any part of the body, heat is withdrawn by the same token. Even if the heart were as hot as glowing iron, it could not warm the hands and feet as it does unless it continually sent new blood to those parts. We also recognize from these considerations that the true purpose of respiration is to bring enough fresh air into the lungs to condense the blood which was rarefied in the right ventricle before it returns to the left; to take blood which has almost been converted into vapor and reconvert it into blood. If this were not done, the blood would not be suitable for the nourishing of the heart's fire. This is confirmed by seeing that animals that have no lungs have only one ventricle, and that unborn children, who cannot use their lungs while enclosed in their mothers' wombs, have an opening through which blood flows directly from the vena cava into the left ventricle, and a tube by which it passes from the vena arteriosa into the aorta without passing through the lungs. Then, how could digestion take place in the stomach if the heart did not send heat there through the arteries, together with some of the most fluid parts of the blood which help to dissolve the food which is placed there? Is it not easy to understand the action which converts the liquid part of these foods into blood if we consider that the blood is distilled possibly more than a hundred or two hundred times each day when passing through the heart? And we need say nothing more to explain nutrition and the production of the several humors of the body, except that the force of the blood, expanding while passing from the heart to the ends of the arteries, brings it about that some of its parts come to rest in certain organs of the body, taking the place of others which they expel; and that certain parts of the blood come to rest in certain places rather than others, according to the location, shape, or size of the pores encountered; just as sieves with holes of different sizes serve to separate different grains from each other. The most remarkable aspect of all this is the production of animal spirits, which are like a very subtle wind, or rather a very pure and lively flame, which continuously rises in great abundance from the heart to the brain, and

thence through the nerves into the muscles, where it produces the movement of all parts of the body. The most agitated and penetrating parts of the blood compose these animal spirits, and no other reason need be sought why these parts go to the brain rather than elsewhere than the fact that the arteries which conduct them are the straightest of all. According to the rules of mechanics, which are the same as the rules of nature, when several objects tend to move toward a place where there is not room enough for all, as is the case when parts of the blood leave the left ventricle and tend toward the brain, the weakest and least agitated of them must be turned aside by the strongest, which thus are the only ones to arrive at their destination.

I had explained all these things in sufficient detail in the treatise which I previously intended to publish. And I continued by showing what the nature of the network of nerves and muscles of the human body must be to enable the animal spirits within to move its members, as one sees when freshly severed heads still move and bite the earth although they are no longer alive. I showed what changes must take place in the brain to cause wakefulness, sleep, and dreams; how light, sounds, odors, tastes, heat, and all the other qualities of external objects can implant various ideas through the medium of the senses; and how hunger, thirst, and the other internal passions are communicated. I explained what must be understood by that animal sense which receives these ideas, by memory which retains them, and by imagination which can change them in various ways and build new ones from them, and thus, distributing the animal spirits in the muscles, move the parts of the body, in response to the objects which are presented to the senses and the passions which are in the body, in as great a variety of ways as our own bodies can move without the guidance of volition. This will hardly seem strange to those who know how many automata or machines can be made by human industry, although these automata employ very few parts in comparison to the large number of bones, muscles, nerves, arteries, veins, and all the other component parts of each animal. Such persons will therefore think of this body as a machine created by the hand of God, and in consequence incomparably better designed and with more admirable movements than any machine that can be invented by man.

Here I paused to show that if there were any machines which had the organs and appearance of a monkey or of some other unreasoning animal, we would have no way of telling that it was not of the same nature as these animals. But if there were a machine which had such a resemblance to our bodies, and imitated our actions as far as is morally possible, there would al. ways be two absolutely certain methods of recognizing that it was still not truly a man. The first is that it could never use words or other signs for the purpose of communicating its thoughts to others, as we do. It is indeed conceivable that a machine could be so made that it would utter words, and even words appropriate to physical acts which cause some change in its

organs; as, for example, if it was touched in some spot that it would ask what you wanted to say to it; if in another, that it would cry that it was hurt, and so on for similar things. But it could never modify its phrases to reply to the sense of whatever was said in its presence, as even the most stupid men can do. The second method of recognition is that, although such machines could do many things as well as, or perhaps even better than, men, they would infallibly fail in certain others, by which we would discover that they did not act by understanding, but only by the disposition of their organs. For while reason is a universal instrument which can be used in all sorts of situations, the organs have to be arranged in a particular way for each particular action. From this it follows that it is morally impossible that there should be enough different devices in a machine to make it behave in all the occurrences of life as our reason makes us behave.

By these two methods we can also recognize the difference between man and animals. For it is a very remarkable thing that there are no men, not even the insane, so dull and stupid that they cannot put words together in a manner to convey their thoughts. On the contrary, there is no other animal, however perfect and fortunately situated it may be, that can do the same. And this is not because they lack the organs, for we see that magpies and parrots can pronounce words as well as we can, and nevertheless cannot speak as we do, that is, in showing that they think what they are saying. On the other hand, even those men born deaf and dumb, lacking the organs which others make use of in speaking, and at least as badly off as the animals in this respect, usually invent for themselves some signs by which they make themselves understood by those who are with them enough to learn their language. And this proves not merely that animals have less reason than men, but that they have none at all, for we see that very little is needed in order to talk. Furthermore, we notice variations among animals of the same species, just as among men, and that some are easier to train than others. It is therefore unbelievable that a monkey or a parrot which was one of the best of its species should not be the equal in this matter of one of the most stupid children, or at least of a child of infirm mind, if their soul were not of a wholly different nature from ours.

Note also that we should not confuse speech with the natural movements that indicate passions, and can be imitated by machines as well as by animals; nor should we think, like some of the ancients, that animals speak although we do not understand their language. For if it were true, they would make themselves understood by us as well as by their fellows, since they have several organs analogous to our own. It is another very remarkable fact that although there are many animals that show more industry than we in some of their behavior, these same animals show none at all in other ways; and so the fact that they do better than we do does not prove that they are rational, for on this basis they would be more rational than any of us, and would surpass us in everything. It proves, on the

contrary, that they are not rational, and that nature makes them behave as they do according to the disposition of their organs; just as a clock, composed only of wheels and springs, can count the hours and measure the time more accurately than we can with all our intelligence.

I then described the rational soul, and showed that it could not possibly be derived from the powers of matter, like the other things I have spoken about, but must have been specially created. I showed also that it would not suffice to place it in the human body, as a pilot in a ship, unless perhaps to move its parts, but that it must be more intimately joined and united with the body in order to have feelings and appetites like ours, and so constitute a real man. For the rest, I elaborated a little on the topic of the soul on account of its great importance; because, next to the error of those who deny God, which I think I have sufficiently refuted, there is none which is so apt to make weak characters stray from the path of virtue as the idea that the souls of animals are of the same nature as our own, and that in consequence we have no more to fear or to hope for after this life than have the flies and ants. Actually, when we know how different they are, we understand more fully the reasons which prove that our soul is by nature entirely independent of the body, and consequently does not have to die with it. Therefore, as long as we see no other causes which might destroy it, we are naturally led to conclude that it is immortal.

PART SIX

SOME PREREQUISITES FOR FURTHER ADVANCES IN THE STUDY OF NATURE

Three years ago, when I had completed the treatise containing all these matters, and when I was beginning to review it for purposes of publication, I learned that people to whom I defer, and whose authority over my actions is hardly less than that of my own reason over my thoughts, had disapproved of a hypothesis in the field of physics that had been published somewhat earlier by another person. I do not want to say that I had accepted that hypothesis, but at least before their censure I could not imagine that it was prejudicial to religion or to the state, and therefore I could see no ground for not professing it if reason convinced me of its truth. This circumstance made me fear that there might be other opinions of mine in which I was misled, despite the great care I had always taken not to accept any new ones which were not very certainly demonstrated, and to write of none that might prove disadvantageous to anyone. This occurrence was enough to make me change my resolution to publish the treatise, for although the reasons for making it were very strong, my inclinations were always much opposed to writing books and I was quick to find other reasons to excuse myself for not publishing. These reasons, on both sides, are such that not

only have I some interest in relating them, but the public may also have some interest in learning them.

I have never entertained any pretensions about the products of my thinking. When the result of the application of my methods was merely my own satisfaction concerning some speculative questions, or perhaps the regulation of my own behavior by the principles which it showed me, I did not feel obliged to write of them For when it comes to morals, everyone is so convinced of his own good sense that there might be as many reformers as individuals if others than those whom God has established as sovereigns over his peoples, or to whom he has given enough grace and zeal to be prophets, were permitted to attempt reforms. So even though my speculations pleased me very much, I believed that other persons had their own speculations which perhaps pleased them even more. As soon, however, as I had achieved some general notions about physics, and when, testing them in various critical problems, I noticed how far they might lead and how they differed from the principles accepted up to this time, I thought that I could not keep them hidden without gravely sinning against the rule that obliges us to promote as far as possible the general good of mankind. For they have satisfied me that it is possible to reach knowledge that will be of much utility in this life; and that instead of the speculative philosophy now taught in the schools we can find a practical one, by which, knowing the nature and behavior of fire, water, air, stars, the heavens, and all the other bodies which surround us, as well as we now understand the different skills of our workers, we can employ these entities for all the purposes for which they are suited, and so make ourselves masters and possessors of nature. This would not only be desirable in bringing about the invention of an infinity of devices to enable us to enjoy the fruits of agriculture and all the wealth of the earth without labor, but even more so in conserving health, the principal good and the basis of all other goods in this life. For the mind is so dependent upon the humors and the condition of the organs of the body that if it is possible to find some way to make men in general wiser and more clever than they have been so far, I believe that it is in medicine that it should be sought. It is true that medicine at present contains little of such great value; but without intending to belittle it, I am sure that everyone, even among those who follow the profession, will admit that everything we know is almost nothing compared with what remains to be discovered, and that we might rid ourselves of an infinity of maladies of body as well as of mind, and perhaps also of the enfeeblement of old age, if we had sufficient understanding of the causes and of all the remedies which nature has provided. It was my intention to devote my whole life to the pursuit of this muchneeded service, and I had found a method which, it seemed to me, should infallibly lead me to it unless I was prevented either by the brevity of life or the paucity of experiments. I judged that the best precaution against these two dangers would be to publish faithfully to the world the little which I had discovered, and to urge men of ability to continue the work by

contributing, each one according to his inclinations and abilities, to the experiments which must be made. I hoped that each one would publish whatever he had learned, so that later investigators could begin where the earlier had left off. In this way mankind would combine the lives and work of many people, and would go much further than any individual could go by himself.

I noticed that experimentation becomes more necessary in proportion as we advance in knowledge. In beginning an investigation it is better to restrict ourselves to our usual experiences, which we cannot ignore if we pay any attention to them at all, than to seek rarer and more abstruse experiences. The reason for this is that these latter are often deceiving when the causes of the more common phenomena are still unknown, as the circumstances on which they depend are almost always so particular and so minute that it is very difficult to discover them. My own procedure has been the following: I first tried to discover the general principles or first causes of all that exists or could exist in the world, without taking any causes into consideration but God as creator, and without using any evidence save certain indications of the truth which we find in our own minds. After that I examined what were the first and commonest effects which could be deduced from these causes; and it seems to me that by this procedure I discovered skies, stars, an earth, and even, on the earth, water, air, fire, minerals, and several other things which are the commonest of all and the most simple, and in consequence the easiest to understand. Then, when I wanted to descend to particulars, it seemed to me that there were so many different kinds that I believed it impossible for the human mind to distinguish the forms or species of objects found on earth from an infinity of others which might have been there if God had so willed. It thus appeared impossible to proceed further deductively, and if we were to understand and make use of things, we would have to discover causes by their effects, and make use of many experiments. In consequence, reviewing in my mind all the objects which had ever been presented to my senses, I believe I can say that I have never noticed anything which I could not explain easily enough by the principles I had found. But I must also admit that the powers of nature are so ample and vast, and that these principles are so simple and so general, that I hardly ever observed a particular effect without immediately recognizing several ways in which it could be deduced. My greatest difficulty usually is to find which of these is the true explanation, and to do this I know no other way than to seek several experiments such that their outcomes would be different according to the choice of hypotheses.

For the rest, I have now reached the point, it seems to me, where I see clearly enough the direction in which we should go in this research; but I also see that the character and the number of experiments required is such that neither my time nor my resources, were they a thousand times greater than they are, would suffice to do them all. In proportion, therefore, to the

opportunity I shall have in the future to do more or fewer of them, I will advance more or less in the understanding of nature. This I expected to convey in my treatise, and I hoped to show so clearly how useful my project might be that I would oblige all those who desire human benefit, all those who are truly virtuous and not merely so in affectation or reputation, both to communicate to me the experiments that they have already made and to assist me in the prosecution of what remained to be done.

But since then other reasons occurred to me which have made me change my mind. I still think that I should continue to write everything that I consider important as soon as I discover its truth, and do so with as much care as if I intended to publish it. In this way I will have additional opportunities to examine my ideas, for doubtless we always scrutinize more closely that which we expect to be read by others than that which we do for ourselves alone, and frequently the ideas which seemed true to me when I first conceived them have appeared false when I wished to put them on paper. Also I would thus lose no opportunity to benefit humanity, if I am capable of it; and if my writings have any value, those into whose hands they fall after my death may use them as may be most appropriate. But I decided that I should never consent to have them published during my life, for fear that the opposition and controversy which they might arouse, and the reputation which they might possibly bring me, would cause me to waste time which I plan to use in research. For although it is true that each man is obligated to do as much as he can for the benefit of others, and that to be of no use to anyone is really to be worthless, yet it is also true that our interest should extend beyond the present time, and that it is well to avoid things which may bring some profit to the living when it is done with the intention of profiting our descendants still more. So I want it to be understood that the little I have learned thus far is a mere nothing compared to what I do not know and yet do not despair of learning. For it is much tile same with those who gradually discover truth in the sciences as with those who, beginning to be rich, find it less difficult to make important acquisitions than they formerly did, when poorer, to make much smaller ones. Or perhaps we should make the comparison with army chieftains, whose forces usually grow in proportion to their victories and who need more skill to maintain themselves after a defeat than they do to win cities and whole provinces after a victory. For to try to conquer all the difficulties and errors which stand in our way when we try to reach the truth is really to engage in battle; and to reach a false conclusion on an important issue is to lose the battle. After such a loss, much more ability is needed to reinstate ourselves in our former position than is required to make great progress when we have already acquired well-tested principles. For myself, if I have thus far found some truths in the sciences, and I trust that the treatises contained in this volume will convince the reader that I have, I can say that these are only the results and consequences of five or six principal difficulties which I have surmounted. These I count as so many battles in

which fortune has been on my side. I would even go so far as to say that I think that two or three further victories of equal importance would enable me to reach my goal, and I am not so old that I cannot look forward to enough leisure, in the ordinary course of nature, for this purpose. But I feel the greater obligation to make good use of the time remaining to me, the more hope I have of being able to do so effectively, and I would no doubt find many occasions to waste time if I published the foundations of my physics. For although my principles are almost all so evident that to hear them is to believe them, and although there are none that I do not believe I can demonstrate, nevertheless, as they could not possibly agree with all the various opinions held by men at large, I foresee that I would often be distracted by the opposition which they would arouse.

One might argue that this opposition would be useful, partly to show me my mistakes, partly so that if there were anything worth while in my ideas, others would learn of it; and as many can see more than one, they would assist me immediately by their insight. But while I recognize that I am extremely likely to make mistakes, and while I rarely have much confidence in the first thoughts that come to me, nevertheless my experience of the objections that may be raised against me does not lead me to expect much profit by them. For I have often been favored with the judgments both of those I took to be friends and of those whom I took to be indifferent, as well as of a few who were moved by malignity and envy to convey what the affection of my friends would conceal. Yet it has rarely happened that an objection was offered which I had not foreseen, except when it was very farfetched; so that I have hardly ever met a critic of my opinions who did not appear to me to be either less rigorous or less equitable than myself. Nor have I noticed that the arguments carried on in the schools have ever brought to light a truth which was previously unknown, for when each person tries to win, he is more concerned to make his views prevail by appearing to be right than he is to weigh the evidence for both sides. Those who have long been good trial lawyers do not therefore make better judges afterwards.

As for the advantages which others might derive from hearing my ideas, they could not be so very great, especially since my ideas are still in a stage where much more has to be done before they can be applied to practice. And I think I may say without vanity that if anyone can accomplish this, it should be myself rather than any other person. Not that there may not be many minds incomparably superior to my own, but that we never understand a thing so well, and make it our own, when we learn it from another as when we have discovered it for ourselves. This is so true in this instance that although I have often explained some of my opinions to very intelligent people, who seemed to understand them very distinctly while I was speaking, nevertheless when they retold them I have noticed that they have almost always so changed them that I could no longer accept them as my own. I should also like to take advantage of this occasion to request posterity

never to believe that any ideas are mine unless I have divulged them myself. I am not at all surprised at the extravagances attributed to the ancient philosophers whose writings we do not possess, nor do I judge in consequence that their ideas were unreasonable. They were the wisest men of their time, so I presume that their ideas have been badly reported. We notice, also, that it has rarely happened that one of their disciples has surpassed them, and I feel sure that the most devoted of the contemporary followers of Aristotle would consider themselves fortunate if they had as much knowledge of nature as he had, even on the condition that they would never have any more. They are like the ivy, which has no tendency to climb higher than the trees which support it, and often grows downward after it has reached the top. For it seems to me that followers also decline; that is, they make themselves somehow less wise than they would be if they abstained from study when they are not satisfied to understand what is intelligibly explained by their author, but insist on finding in him the solutions of many problems of which he says nothing, and of which he has perhaps never thought. Just the same, their manner of philosophizing is very convenient for those who have only very mediocre minds; for the obscurity of the distinctions and principles which they use enables them to talk of all things as bravely as though they understood them, and to defend all they say against the deepest and cleverest thinkers, as there is no way to convince them. In this they appear to me similar to a blind man who wishes to fight on even terms with one who can see, and so brings him to the back of some very dark cave. These people, I may say, are interested in my abstaining from the publication of my principles of philosophy; for since these are very simple and evident, I would be doing much the same to them as though I opened some windows and let the light of day enter into that cave where they had retired to fight. But even the best minds need not wish to know my principles; for if they want to be able to talk about all things and gain the reputation of being learned, they can accomplish this more easily by being satisfied with the appearance of truth, which can be found without much trouble in all sorts of matters, than by seeking truth itself. Truth can be discovered only little by little, and in a few subjects, so that he who pursues truth is often obliged to admit his ignorance when discussing a subject which he has not investigated. But if they prefer the knowledge of a little truth to the vanity of seeming to know everything, as is no doubt preferable, and if they wish to pursue a plan similar to mine, it is not necessary for me to tell them anything more than I have already said in this Discourse, for if they are capable of going further than I have gone, they will be still more capable of finding for themselves everything which I think I have found. This is especially true because I have always proceeded in a natural order, so that it is certain that what remains to be discovered is more difficult in itself and more recondite than what I have so far encountered. They would also experience much less pleasure in learning it from me than in discovering it for themselves. In addition, they would thus acquire the habit of discovery by seeking easy things first, and pass

gradually and by degrees to more difficult ones-a habit which will prove much more useful than all my information could possibly be. As for myself, I am persuaded that if I had been taught in my youth all the truths of which I have since sought demonstrations, and if I could have learned them without difficulty, I might never have learned any others, or at least, I would never have acquired the habit and ability that I believe I possess, always to find new truths in proportion to the efforts I make to find them. In a word, if there is one task in the world that cannot be finished as well by another as by the one who started it, it is this one at which I am working.

It is true that as far as the related experiments are concerned, one man is not enough to do them all; but he could not usefully employ other hands than his own, unless those of workers or other persons whom he could pay. Such people would do, in the hope of gain, which is a very effective motive, precisely what they were told. As for those volunteers who might offer to do it out of curiosity or the desire to learn, besides the fact that ordinarily they are stronger in promises than in performance and that they make nothing but beautiful proposals of which none ever succeeds, they would infallibly expect to be paid by the explanation of some difficulties, or at least in compliments and useless conversation, which would necessarily consume so much of the time needed for investigation that the assistance would be at a net loss. As for the experiments which others have already made, even if they were willing to communicate them, which those who call them secrets never do, they are for the most part so complicated with unneeded details and superfluous ingredients that it would be very difficult for the investigator to discover their core of truth. Besides this, he would find almost all these experiments so poorly explained or even false, because those who performed them forced themselves to make them appear conformable to their principles, that if some of them were useful, they could not counterbalance the time that would be lost in picking them out. So even if there were someone in the world who could be recognized without question as capable of making the greatest and the most beneficial discoveries, and even if in consequence all other men attempted by every means to aid him in the accomplishment of his designs, I do not see that they could do anything except contribute to the costs of tile necessary experiments, and see that his leisure is not interrupted by the importunities of anyone. But I am not so presumptuous as to promise anything extraordinary, nor do I indulge in such vain fancies as to imagine that the public ought to be much interested in my plans. Finally, I am not so base in spirit that I would be willing to accept from anyone whomsoever a favor which it might be thought I had not deserved.

All these considerations taken together made me decide, three years ago, that I did not wish to publish the treatise which I had on hand, and I even resolved never during my lifetime to permit others to read any paper of such a general nature that they might understand the foundations of my

physics. But two other reasons have occurred since which have obliged me to submit herewith some detailed essays, and to give the public some account of my doings and my plans. The first reason is that if I did not do so, several people who knew of my previous intention to publish several essays might suppose that my reasons for abstaining were less honorable than they really are. For although I do not care too greatly for reputation-I might even say that I dislike it insofar as I consider it destructive of peace of mind, which I esteem above all things—nevertheless I have never tried to hide my actions as though they were criminal. Neither have I made much effort to remain unknown, partly because I would have thought I was doing myself an injustice, partly because that would have produced a certain disquiet unfavorable to that perfect peace of mind which I desire. I have always tried to remain indifferent to having or not having a reputation, but since I could not avoid having some kind, I thought I should at least do my best to avoid a bad one. The other reason which has obliged me to write this is that I observe a constantly greater retardation in my plan to enlighten myself, because of an infinity of experiments which I must do, so that it is impossible for me to succeed without the aid of others. And although I do not flatter myself enough to hope that the public would be much interested in what I am doing, nevertheless I do not wish to be so remiss in upholding my own interests as to give occasion to those who survive me to reproach me, some day, on the ground that I might have accomplished many much better things than I did if I had not been too negligent to explain how others could contribute to my designs.

And I thought that it would be easy to choose some topics which would not be too controversial, which would not force me to divulge more of my principles than I wish to, and which would demonstrate clearly enough what I could or could not do in the sciences. It is not for me to say whether I have succeeded, and I do not wish to influence anyone's decisions by speaking of my own writings; nevertheless I should like to request the reader to examine them. In order to add to the opportunities of judging, I also request all who find some objections to my ideas to take the trouble to send them to my publisher. He will inform me, and I shall try to have my replies published at the same time as the objections. By this means the reader, seeing both together, will more easily judge of the truth. For I do not promise ever to make lengthy replies, but only to admit my mistakes very frankly if I recognize them; or, if I cannot recognize them, to say simply what I believe to be required for the defense of what I have written. But I shall not go on to explain any new material, for fear of engaging in an endless chain of tasks.

If some of the matters I deal with at the beginning of Optics and Meteorology should at first sight appear offensive, because I call them hypotheses and do not try to prove them, let the reader have the patience to read all of it with attention, and I hope that he will be satisfied with the

result. For it seems to me that the arguments follow one another in such a way that, just as the last principles are demonstrated by the first ones which are their causes, so these first ones are reciprocally demonstrated by the last which are their effects. And one must not suppose that I have here committed the fallacy which logicians call circular reasoning; for as experience makes most of the effects very certain, the causes from which I deduce them serve not so much to prove as to explain them-on the contrary, the truth of the hypotheses is proved by the actuality of the effects. And I have called them hypotheses only to let it be known that although I think I can deduce them from the first truths which I have previously explained, I expressly desired not to make the deduction. For there are certain people who imagine that they can learn in one day all that another has thought in twenty years, as soon as he has only spoken two or three words, and who are only the more subject to err and less capable of truth as they are more penetrating and lively of spirit. I should like to prevent these people from building some extravagant philosophy on what they believe to be my principles, for the fault might be attributed to me. As for my real opinions, I do not apologize for their novelty, especially since I am sure that anyone who attends to the argument will find them so simple and so conformable to common sense that they will seem less extraordinary and strange than any other opinions that can be held on the same subjects. I do not claim, either, that I am original in any of these ideas, but only that I have never accepted them because they were maintained by others, nor because they were not so maintained, but only because reason persuaded me of their truth.

And if the invention described in Optics cannot immediately be built, I do not think it is therefore faulty. Since much skill and practice are necessary in order to make and adjust the machines which I have described, without their having any defects, I would not be less astonished to find it successful on the first attempt than I would be if someone could learn to play the lute excellently in a single day, for the sole reason that he had been given some excellent sheet music. And if I write in French, which is the language of my country, rather than in Latin, which is that of my teachers, it is because I hope that those who rely purely on their natural intelligence will be better judges of my views than those who believe only what they find in the writings of antiquity. And those who combine good sense with studiousness, whom alone I wish for my judges, will not, I am sure, be so partial to Latin that they will refuse to accept my reasons because I explain them in the vulgar tongue.

For the rest, I do not wish to speak here in detail of the progress in the sciences which I hope to make in the future, nor to commit myself to any promise to the public which I am not sure of fulfilling. I shall therefore only say that I have resolved to employ as much of my life as remains wholly in trying to acquire some knowledge of nature, of such a sort that we may derive rules of medicine more certain than those which we have had up to

the present. My inclinations are so far removed from any other plans, especially those which can be useful to some only by harming others, that if circumstances forced me to employ those plans, I do not think I would be capable of carrying them to a successful conclusion. The declaration I am here making will not, I well know, procure me any worldly advantages, but I have no desire for them; and I shall always consider myself more obligated to those by whose favor I shall enjoy uninterrupted leisure than I would be to those who offered me the most honorable office on earth.